

The Two-Body Problem in the Context of Lagrangians

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Introduction

This is a brief technical note on how one may use the Lagrangian method to simplify the two-body problem, enabling it to be solved as if there were only one particle. As always, email me should you find an incorrect line of working or have suggestions that can make this article more pedagogically sound:

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1 Reduced Mass in the Lagrangian

Recall that the (non-relativistic) Lagrangian is defined as

$$\mathcal{L} = T - V,$$

where T and V denote the total kinetic and potential energies of the system, respectively.

Now, consider two particles, masses m_1 and m_2 , with respective position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$. We assume that the potential energy of the two objects depends only on their distance.

The Lagrangian is then

$$\mathcal{L} = \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 - V(|\mathbf{r}_2 - \mathbf{r}_1|). \quad (1)$$

Now, recall that the system's center of mass is given by the position vector

$$\mathbf{r}_{\text{CM}} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}. \quad (2)$$

If we take the origin at the center of mass, that would imply $\mathbf{r}_{\text{CM}} = \mathbf{0}$, so that we obtain

$$m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2 = \mathbf{0}. \quad (3)$$

from Equation 2.

Moreover, let's define a new vector

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1. \quad (4)$$

Notice the length of $\mathbf{r}$ is equal to $|\mathbf{r}_2 - \mathbf{r}_1|$.

Now, from Equation 3, we have

$$\mathbf{r}_1 = -\frac{m_2}{m_1} \mathbf{r}_2.$$

Thus,

$$\mathbf{r}_1 = -\frac{m_2}{m_1} (\mathbf{r} + \mathbf{r}_1),$$

$$\begin{aligned} \left(1 + \frac{m_2}{m_1}\right) \mathbf{r}_1 &= \frac{m_1 + m_2}{m_1} \mathbf{r}_1 \\ &= -\frac{m_2}{m_1} \mathbf{r}, \end{aligned}$$

where the last equality holds by the equation directly above. This gives

$$\mathbf{r}_1 = -\frac{m_2}{m_1 + m_2} \mathbf{r}. \quad (5)$$

By a similar argument, one can show that

$$\mathbf{r}_2 = \frac{m_1}{m_1 + m_2} \mathbf{r}. \quad (6)$$

We substitute Equations 5 and 6 into the Lagrangian given in Equation 1:

$$\begin{aligned}
\mathcal{L} &= \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 - V(|\mathbf{r}_2 - \mathbf{r}_1|) \\
&= \frac{1}{2}m_1 \left(\frac{m_2}{m_1 + m_2} \dot{\mathbf{r}} \right)^2 + \frac{1}{2}m_2 \left(\frac{m_1}{m_1 + m_2} \dot{\mathbf{r}} \right)^2 - V(r) \\
&= \frac{1}{2} \left(\frac{m_1 m_2^2}{(m_1 + m_2)^2} + \frac{m_2 m_1^2}{(m_1 + m_2)^2} \right) \dot{\mathbf{r}}^2 - V(r) \\
&= \frac{1}{2} \left(\frac{m_1 m_2^2 + m_2 m_1^2}{(m_1 + m_2)^2} \right) \dot{\mathbf{r}}^2 - V(r) \\
&= \frac{1}{2} \left(\frac{m_1 m_2 (m_1 + m_2)}{(m_1 + m_2)^2} \right) \dot{\mathbf{r}}^2 - V(r) \\
&= \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) \dot{\mathbf{r}}^2 - V(r) \\
&= \frac{1}{2} \mu \dot{\mathbf{r}}^2 - V(r),
\end{aligned}$$

where $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is the *reduced mass* of the system.

2 Applications

One fun application of the formula above is in the problem of two masses, m_1 and m_2 , joined together by a spring of force constant k .

We define coordinates so that $\mathbf{r}_1$ and $\mathbf{r}_2$ denote the displacements of each mass from the spring's *equilibrium extension*, allowing us to disregard the spring's length in the Lagrangian.

We use the expression of the Lagrangian in reduced mass, as derived in Section 1, to obtain

$$\mathcal{L} = \frac{1}{2} \mu \dot{\mathbf{r}}^2 - \frac{1}{2} k \mathbf{r}^2,$$

where of course $\mathbf{r} := \mathbf{r}_2 - \mathbf{r}_1$.

Now,

$$\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}} = \mu \dot{\mathbf{r}}, \tag{7}$$

and

$$\frac{\partial \mathcal{L}}{\partial \mathbf{r}} = -k \mathbf{r}. \tag{8}$$

The Euler-Lagrange Equation implies

$$\frac{\partial \mathcal{L}}{\partial \mathbf{r}} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}} \right) = 0,$$

so substituting our findings in Equations 7 and 8 above gives

$$-k\mathbf{r} - \mu\ddot{\mathbf{r}} = 0,$$

or

$$\ddot{\mathbf{r}} + \frac{k}{\mu}\mathbf{r} = 0,$$

which is the familiar harmonic differential equation¹. We therefore have

$$\omega = 2\pi f = \sqrt{\frac{k}{\mu}},$$

which yields

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}.$$

The equation above is important in IR spectroscopy, a vital analytical tool in organic chemistry[Cla+12].

¹A more detailed discussion of equations of this form in oscillations is contained in Chapter 4 of [Mor08]. One may also refer to Chapter 1 of [SS03]

References

- [Cla+12] Jonathan Clayden et al. *Organic Chemistry*. 2nd ed. Oxford University Press, 2012.
- [Mor08] David Morin. *Introduction to Classical Mechanics: With Problems and Solutions*. Cambridge University Press, 2008.
- [SS03] Elias M. Stein and Rami Shakarchi. *Fourier Analysis: An Introduction*. Vol. 1. Princeton Lectures in Analysis. Princeton University Press, 2003.